

Supplementary Material 4: Scripts

Folder	Script	Description
script	additional_functions.R	Helper functions to render the main paper.
script	evaluation.py	Python script implementing additional evaluation metrics such as RTA and GS.
script	nldr_code.R	Wrapper functions for running multiple NLDR methods (UMAP, tSNE, PHATE, PaCMAP, TriMAP) with different parameters.
two_nonlinear	01_gen_data.R	Generates the 2NC7 dataset.
two_nonlinear	02_gen_true_model.R	Creates the true structure of 2NC7 data.
two_nonlinear	03_gen_embeddings.R	Computes multiple NLDR embeddings for the 2NC7 data.

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Folder	Script	Description
two_nonlinear	04_gen_mse_for_diff_methods.R	Computes HBE with varying bin widths (a_1) for all NLDR embeddings.
two_nonlinear	05_gen_rm_lwd_mse.R	Computes HBE with varying low density bin cutoff for all three binwidth (a_1) choices.
two_nonlinear	06_gen_model_with_tSNE.R	Fits the model for the layout a.
two_nonlinear	07_example_evaluation_metrics.R	Calculates evaluation metrics for all NLDR layouts.
two_nonlinear	08_gen_model_with_PHATE.R	Fits the model for the layout c.
five_gau_clusters	01_five_gaussian_cluster_data_emb.R	Generates data and multiple NLDR embeddings.
five_gau_clusters	02_gen_model_with_tSNE.R	Fits the model for the layout a.
five_gau_clusters	03_gen_model_with_UMAP.R	Fits the model for the layout b.
five_gau_clusters	04_gen_model_with_PaCMAP.R	Fits the model for the layout c.
c_shaped_dens_str	01_gen_data.R	Generates the 2-D curved sheet dataset.
c_shaped_dens_str	02_gen_embeddings_uni_dens.R	Generates multiple NLDR embeddings.

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Folder	Script	Description
c_shaped_dens_str	03_gen_model_with_tSNE.R	Fits the model for the tSNE layout.
pbmc3k	01_obtain_pca_author.R	Obtains author's PCA results.
pbmc3k	02_obtain_umap_authors.R	Obtains author's UMAP embeddings.
pbmc3k	03_gen_umap_diff_param.R	Generates multiple UMAP embeddings with different hyperparameter values.
pbmc3k	04_gen_tsne_diff_param.R	Generates multiple tSNE embeddings with different hyperparameter values.
pbmc3k	05_gen_phate.R	Generates a PHATE embeddings with default hyperparameters.
pbmc3k	06_gen_trimap.R	Generates a TriMAP embeddings with default hyperparameters.
pbmc3k	07_gen_pacmap.R	Generates a PaCMAP embeddings with default hyperparameters.

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Folder	Script	Description
pbmc3k	08_gen_mse_for_diff_methods.R	Computes HBE with varying bin widths (a_1) for all NLDR embeddings.
pbmc3k	09_gen_scDEED.R	Generates UMAP embeddings from scDEED results.
pbmc3k	10_pre_process_for_embedding.R	Generates PBMC3k data used for scDEED results.
pbmc3k	11_gen_mse_for_diff_tsne_scD.R	Computes HBE with varying bin widths (a_1) for tSNE embeddings.
pbmc3k	12_gen_mse_for_diff_umap_scD.R	Computes HBE with varying bin widths (a_1) for UMAP embeddings.
pbmc3k	13_gen_model_with_UMAP.R	Fits the model for the layout a.
pbmc3k	14_gen_model_with_tSNE.R	Fits the model for the layout e.
pbmc3k	15_gen_model_with_UMAP_scD.R	Fits the model for the layout a.
pbmc3k	16_gen_model_with_tSNE_scD.R	Fits the model for the layout b.
pbmc3k	17_evaluation_metrics.R	Calculates evaluation metrics for all NLDR layouts.

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Folder	Script	Description
pbmc3k	18_evaluation_metrics_scD.R	Calculates evaluation metrics for all NLDR layouts.
mnist	01_data_preprocessing.R	Computes first 10 principal components and save data.
mnist	02_gen_diff_embeddings.R	Generates multiple NLDR embeddings.
mnist	03_gen_mse_for_diff_methods.R	Computes HBE with varying bin widths (a_1) for all NLDR embeddings.
mnist	04_gen_model_with_tSNE.R	Fits the model for the layout a.
mnist	05_evaluation_metrics.R	Calculates evaluation metrics for all NLDR layouts.
mnist	06_link_brush_layout_e.R	Creates interactive linked brushing with layout e.

Table 1: R and Python script files used to generate outputs in the main paper.